Year 8
Mathematics

Unit 2: Equations and Formulae

2.1 Solving One-Step Equations

Solving One-Step Equations

STARTER · 5 MIN

Date: _____ **Title: Solving One-Step Equations**

LEARNING OBJECTIVES

- To be able to write and solve simple equations

STARTER *Answer in your book. No calculators.*

1. What is the inverse of + 5?
2. What is the inverse of × 3?
3. □ + 7 = 12. Find □.
4. 4 × □ = 36. Find □.

Starter — Answers

STARTER

Tick or correct in green pen

1. **– 5**
2. **÷ 3**
3. **5**
4. **9**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can use inverse operations to solve one-step equations	☐
SILVER	I can form and solve more complex equations	☐
GOLD	I can write a one-step equation from a word problem and solve it correctly	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Solving One-Step Equations

INTRODUCTION · 3 MIN

An equation has a letter and an = sign. Solving it means finding the letter.

Inverse

Reverses an operation.

$+ \leftrightarrow -$ $\times \leftrightarrow \div$

Balance

Do the same to both sides.

$x + 6 = 12 \rightarrow -6$ both sides

Solve

Get the letter on its own.

$x = 6$

Check

Substitute back in.

$6 + 6 = 12$, correct

Examples 1 and 2 — Adding and Subtracting

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint + ↔ − Do the same to both sides

Example 1

Solve

$x + 6 = 12$

Example 2

Solve

$x - 9 = 3$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Inverse of + 6 is − 6

$x + 6 = 12$

$x = 12 - 6$

$\mathbf{= 6}$

Tip: check: 6 + 6 = 12.

Example 2

Inverse of − 9 is + 9

$x - 9 = 3$

$x = 3 + 9$

$\mathbf{= 12}$

Tip: not 3 − 9.

Task 1

PRACTICE · 9 MIN

BRONZE *Solve*

Hint + ↔ − Do the same to both sides

1. $x + 4 = 15$

2. $m - 3 = 5$

3. $t + 9 = 14$

4. $m - 6 = 8$

5. $12 + y = 20$

★ $p - 7 = -2$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 $x = 11$

2 $m = 8$

3 $t = 5$

4 $m = 14$

5 $y = 8$

★ $p = 5$

Examples 3 and 4 — Multiplying and Dividing

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint $5x = 20$: $\div 5$ $\quad$ $d \div 4 = 10$: $\times 4$

Example 3

Solve

$5x = 20$

Example 4

Solve

$\frac{d}{4} = 10$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

5x means 5 × x: divide by 5

$$5x = 20$$
$$x = 20 \div 5$$
$$\mathbf{= 4}$$

Tip: not 20 × 5.

Example 4

Inverse of ÷ 4 is × 4

$$\frac{d}{4} = 10$$
$$d = 10 \times 4$$
$$\mathbf{= 40}$$

Tip: check: 40 ÷ 4 = 10.

Task 2

PRACTICE · 9 MIN

SILVER *Solve*

Hint $5x = 20$: ÷ 5 $d \div 4 = 10$: × 4

1. $4x = 28$

2. $3R = 6$

3. $\frac{d}{5} = 8$

4. $7k = 63$

5. $\frac{a}{3} = 12$

★ $2y = 9$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $x = 7$

2 $R = 2$

3 $d = 40$

4 $k = 9$

5 $a = 36$

★ $y = 4.5$

Examples 5 and 6 — Writing Equations

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint Choose a letter Check by substituting

Example 5

(2 marks)

I think of a number, add 15.

The answer is 42.

Write an equation and solve it.

Example 6

(2 marks)

5 identical books cost $35.

Write an equation for the cost b of one book. Solve it.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Call the number *n*

$$n + 15 = 42$$
$$n = 42 - 15$$
$$\mathbf{= 27}$$

Tip: check: 27 + 15 = 42.

Example 6

5 books at *b* dollars each

$$5b = 35$$
$$b = 35 \div 5$$
$$\mathbf{= \$7}$$

Tip: say what the letter stands for.

Task 3

PRACTICE · 9 MIN

GOLD *Exam-style: write and solve*

Hint Choose a letter Check by substituting

1. A number minus 8 is 13.
   Find the number.

2. 6 times a number is 54.
   Find the number.

3. A number divided by 7 is 4.
   Find the number.

4. Sam is 4 years older than Ali.
   Sam is 17. How old is Ali?

5. A square has perimeter 36 cm.
   Find its side s.

★ Spot the mistake:
$5x = 20$, so $x = 100$

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. $n - 8 = 13$: $n = 21$

2. $6n = 54$: $n = 9$

3. $\frac{n}{7} = 4$: $n = 28$

4. $a + 4 = 17$: $a = 13$

5. $4s = 36$: $s = 9$ cm

★ **Divide, not multiply:** $x = 4$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Spot the mistake: $x + 6 = 12$, so $x = 12 + 6 = 18$

SILVER Spot the mistake: $x - 9 = 3$, so $x = 3 - 9 = -6$

GOLD Spot the mistake: $4x = 28$, so $x = 28 + 4 = 32$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $x = 12 - 6 = 6$

SILVER $x = 3 + 9 = 12$

GOLD $x = 28 \div 4 = 7$

Exam Practice — One-Step Equations

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint + ↔ − Do the same to both sides

Question 1

(2 marks)

Solve

(a) $x + 7 = 19$ (1)

$x =$ .....

(b) $y - 4 = 11$ (1)

$y =$ .....

Question 2

(2 marks)

Solve

(a) $6m = 42$ (1)

$m =$ .....

(b) $\frac{k}{3} = 5$ (1)

$k =$ .....

Exam Practice — Equations

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint $5x = 20$: ÷ 5 $d \div 4 = 10$: × 4

Question 3

(2 marks)

Solve $4x = 18$

$x =$ .....

Question 4

(2 marks)

Solve $w - 2.5 = 7.5$

$w =$ .....

Exam Practice — Forming Equations

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint Choose a letter Check by substituting

Question 5

(2 marks)

The perimeter of an equilateral triangle is 27 cm.

Write an equation for the side length s and solve it.

..... cm

Question 6

(2 marks)

Lina saves \$$x$ a week.
After 8 weeks she has \$96.

Write an equation and find x.

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 12 — B1

(b) 15 — B1

Tip: use the inverse.

Question 2

(a) 7 — B1

(b) 15 — B1

Tip: ÷ 3 undone by × 3.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$x = 18 \div 4$ M1

$= 4.5$ A1

Tip: answers need not be whole.

Question 4

$w = 7.5 + 2.5$ M1

$= 10$ A1

Tip: inverse of − is +.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$3s = 27$ — M1

$s = 9$ cm — A1

Tip: equilateral: 3 equal sides.

Question 6

$8x = 96$ — M1

$x = 12$ — A1

Tip: check: 8 × 12 = 96.